

Immunoglobulin for Therapeutics



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1. The current progress

Immunoglobulin is a globulin with antibody activity or chemical structure that is similar to an antibody molecule. Immunoglobulin is a four-peptide chain structure consisting of two identical light chains and two identical heavy chains connected by disulfide bonds between the chains. There are five types of immunoglobulin: immunoglobulin G (IgG), immunoglobulin A (IgA), immunoglobulin M (IgM), immunoglobulin D (IgD), and immunoglobulin E (IgE). The plasma cells transformed from B lymphocytes secrete antibodies, and each B lymphocyte line can produce only one kind of its own antibodies, specific to a specific antigen epitope. The antibodies produced from a single cell line are called monoclonal antibodies. In recent decades, monoclonal antibody drugs have gradually occupied an important position in the field of biotechnology pharmaceuticals and become the main development direction in the field of biomedicine. For the global market, antibody drugs become the focus of the international pharmaceutical industry to fight for share. Monoclonal antibody therapy mainly uses its targeting to intervene various pathways in the process of disease occurrence and development or to activate the host's immunity to diseases. [1]

Currently available monoclonal antibody drugs generally fall into the following categories:

1. Monoclonal antibody agents for the treatment of diseases (especially for tumors).
2. Antitumor monoclonal antibody conjugates.
3. Monoclonal antibodies for the treatment of other diseases.

Monoclonal antibody agents usually target disease-related antigens or specific receptors on the cell surface. For example, rituximab, a monoclonal antibody drug, was first approved by the us FDA for the treatment of tumors. Antitumor effects of monoclonal antibody coupling, or called immune coupling (Immunoconjugate) by monoclonal antibody with treatment function of material (such as: radioactive nuclide, toxin and drug, etc.) of two parts, including radiation immune coupling, poison, chemical coupling, in addition to the enzyme combined with monoclonal antibody coupling material, photosensitizer combined with monoclonal antibody coupling.

2. Trend :

Therapeutic and diagnostic antibodies have become the fastest growing field in biopharmaceutical applications. There are currently 18 monoclonal antibodies on the market and more than 100 in clinical trials, highlighting the importance of these therapies and advances in antibody engineering. In addition, engineered antibodies are expected to be the source of more

than a third of biotechnology revenues by 2008. [2] With advances in technology, antibodies used for therapeutic and prophylactic indications have been transferred from complete mice to partial mice (mainly humans) and then to complete humans. The huge advances in production that have enabled us to produce monoclonal antibodies on a scale that can provide enough antibody drugs and save some of the cost have made this area very attractive for pharmaceutical companies. An important consequence of these advances is that antibody drugs have greater potential for use in chronic environments, greatly expanding their use in biopharmaceutical applications.[3] The body produces several antibodies that seem to perform specific functions under different conditions and at different anatomical sites. Most of the currently approved antibody-based drugs are in the IgG1 class and glycation occurs in the Fc(fragment crystallizable) domain. Glycosylation has been shown to be involved in antibody-mediated cell-specific response events: antibody dependent cell-mediated cytotoxicity (ADCC) and receptor-recognition event-mediated complement dependent cytotoxicity (CDCC). Monosaccharide forms of antibodies can improve the efficacy of certain clinical outcomes. In general, IgG1 and IgG3 antibodies are effective in initiating ADCC, while IgG2 is easy to repair complement, but poor in inducing ADCC. IgG4 appears to be inefficient on both ADCC and CDCC. Interestingly, antibodies can be redesigned to increase their ability to recruit complement, or to modulate their affinity to various types of Fc receptors, thereby performing the function of ADCC through mutations in specific amino acids in the Fc region. With advances in analytical techniques, antibodies are generally susceptible to some degree of lyophilization and destruction, as evidenced by the formation of large insoluble aggregates. Recent studies have shown that freeze-drying antibodies that target a specific proportion of residual water (typically between 1% and 8%) achieve optimal stability in the dry state and remain stable during reconstruction. Higher water content in lyophilized antibody preparations is associated with increased aggregation (as shown by SEC) and asparagine deamidation and isomerization (as shown by HIC). Additional studies were conducted to determine the optimal freeze-drying process, in which the secondary drying phase was performed at higher temperatures. This technique optimizes the moisture content of the freeze-dried cake, making the final product structurally stable (as measured by DSC) during extended storage. Solid-state structure studies can be used to monitor conformational changes and aggregation events during the preparation of lyophilized monoclonal antibodies. Improving the protogenic structure and reducing polymerization can obtain sufficient amounts of carbohydrate excipients doped to fulfill the hydrogen bond protein surface requirements, presenting a key role for nonwater molecules as placeholders for water molecules in a dry, in a dry state. [4] The fact that the addition of excessive carbohydrates does not improve the stability of the excipients again confirms the conclusion that the limited number of sites for antibodies must be protected in this way. Various carbohydrates or polyols, such as sucrose, trehalose, and mannitol, have been shown to have this stabilizing effect. The rehydration rate is

an important parameter in the preparation of lyophilization. [5] When water is removed at different drying stages of the freeze-drying cycle and replaced by a layer of non-water excipients, antibody drugs can undergo structural changes leading to unnatural conformations. When water replaces non-water excipients, recombination usually provides satisfactory results for antibodies if the rehydration is slow enough to restore the natural conformation. However, if the antibody does not have sufficient time or ability to regain its initial state during recombination, widespread aggregation may result. Although both mannitol and trehalose stabilize human IgG during the spray freeze-drying process, dual emulsion solvent evaporation of spray freeze-drying human IgG into PLGA microspheres has been shown to produce extensive antibody aggregation. This result is likely the result of the thermodynamic stress experienced at the water/organic solvent interface, a concern that can be mitigated by a variety of processes and excipients.

3. Challenges :

Due to the nature of its proteins, the general formulation of antibodies is similar to that of most protein therapies. In general, proteins can be degraded when exposed to high and low temperatures, light, pH, agitation, shear stress, certain metals, and organic solvents. The ability to evaluate the physicochemical and thermodynamic instability of antibody drugs with advanced analytical techniques has also led to the identification of potential events that further distinguish proteins from this particular unique property: changes in Fc glycosylation, partial c-terminal lysine treatment, Fc methionine oxidation, cleavage in the hinge region, and glycosylated lysine residues. In addition to these problems, it is difficult to prepare dosing materials with a protein concentration of 100 mg/ml or higher. Because early preclinical evaluation is critical to the successful identification of new biologic therapeutic products, common preformulation strategies are often used in the preliminary study of new antibody drugs. In studies using related animal models, so the initial formulation may use isotonic, non-hemolytic, slightly acidic solutions that can be administered intravenously or subcutaneously.

A complete full-length antibody consists of several segments with natural bending sites. One of these sites is at the bending point between the Fc and Fv domains. Cleavage in the hinge region is a common concern in evaluating the stability of antibody drugs. [6] Molecular classification methods such as sodium dodecyl sulfate - polyacrylamide gel electrophoresis (SDS-PAGE) and screening high performance liquid chromatography (SE-HPLC) can be used to rapidly identify cleavage events in antibody drugs. Fragmentation of full-length antibody drugs is a common phenomenon and inevitably affects their function. In general, cleavage forms exist in such low quantities that few changes in its potency will be seen. At the very least, there may be some slight reduction in the efficacy of antibody drugs after fragmentation, or they may be

more prone to protease, degradation, and clearance events, limiting the duration of their efficacy. In the worst case, the preparation of fragmented antibodies may lead to different biological distributions, leading to different safety. This may be of particular concern if antibodies are used to deliver cytotoxic chemotherapies. Fortunately, the fragmentation of antibodies can easily be limited to the formulation product. Although liquid antibody preparations are cheaper, faster to develop, and generally easier to prepare for administration than other dosage forms, liquid antibody preparations are prone to oxidation, deamination, aggregation, and fragmentation. In these events, water is the common culprit: water mediate electron transfer during oxidation and deamination, and its addition to peptide bonds may be a key step in fragmentation. The thermodynamic stress that leads to protein aggregation is due to hydrophobic protein surfaces exposed to water and attempts to find lower energy states through unnatural protein-protein interactions. Therefore, water is a big problem for stabilizing antibody-based drugs. Freeze-drying or introduction of hydrophobic polymer systems can reduce the effect of water on the formulation of antibody drugs.

The entire spectrum of antibody fragments has been identified through protein engineering studies, many of which show significant potential for unique and novel clinical opportunities. These smaller recombinant antibody fragments, monovalent antibody fragments, and engineered variants, diabodies, triabodies, minibodies, and single-domain antibodies, which are less expensive to manufacture but still retain the targeted specificity of full-length antibodies, are an emerging form of antibody therapy. In fact, some monoclonal antibodies may be more effective and more widely used than full monoclonal antibodies. This new subset of antibodies can be expected to present new formulation challenges. Individual V domains have been found to be poorly soluble and tend to aggregate, but these problems can be overcome by identifying mutants that minimize hydrophobic interfaces and selecting through phage displays. In general, binding of antibodies to effector proteins does not necessarily inactivate the function of the target protein. However, when antibodies are effectively inactivated, Fab fragments without Fc domains can often act as potent binders, shielding binding sites or enzyme elements on large molecules. For example, the Fab fragment extracted from sheep antivenin antibody (Digibind, from GSK) is an effective antivenin. Storing liquid formulations of such antivenoms in subtropical countries that cannot be refrigerated is a major recipe challenge. While lyophilized products performed well under these conditions, the pH 4.0 (acetate buffer) formulation was also found to be stable and effective at room temperature for at least 1 year, suggesting that the material would be used for room temperature characteristics in tropical and subtropical regions. Notably, this stable formulation does not require the presence of protective carbohydrate excipients. To improve drug delivery, antibody fragments are incorporated into many types of molecules, such as radionuclides, cytotoxic drugs, toxins, peptides and proteins, enzymes, and liposomes.

Immunoliposomes, consisting of monoclonal antibody fragments, are reported to deliver drugs to the brain, enabling molecules to cross the powerful barrier of the blood-brain barrier.

References :

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